

SMART FARM
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Converting a Traditional Farm into an AI/ML-Powered Automated Smart Farm

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Our Team → R26 – IT - 093



IT22139894
Shamika G.L.
(Group Leader)



IT22282118
Illangasinhe I.M.A.D.



IT22136138
Jayawardhana H.A.S.



IT22134608
Arachchi W.A.A.I.W.

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Problem & Motivation

Context

Traditional livestock farming still depends heavily on manual observation and experience for feeding, animal health monitoring, reproduction management, and market decisions.



Observed Problems

- Inconsistent feeding practices and difficulty in providing personalized nutrition.
- Delayed detection of diseases and abnormal animal behavior.
- Difficulty in predicting healthy births and successful egg hatching.
- Limited real-time information about livestock market prices and demand.
- Manual farm management reduces efficiency and increases operational costs.
- Lack of one integrated intelligent platform for livestock management.

Solution Overview

An **integrated AI/ML-powered smart livestock management platform** that transforms traditional farm operations through intelligent monitoring, prediction, automation, and decision support.

Birth & Egg Hatch Prediction

- Analyze image, sensor and historical data.
- Predict healthy calf birth or successful egg hatching.

Disease Detection & Emergency Alerts

- Monitor animal behavior using cameras and sensors.
- Detect abnormal behavior and early disease symptoms.
- Notify farmers or veterinarians.

Personalized Animal Feeding

- Identify animal type, size and health condition.
- Recommend suitable feed quantity and Nutrition.

AI-Based Market Prediction

- Analyze historical and real-time market information.
- Forecast livestock/product prices and demand.

Non-Functional Goals

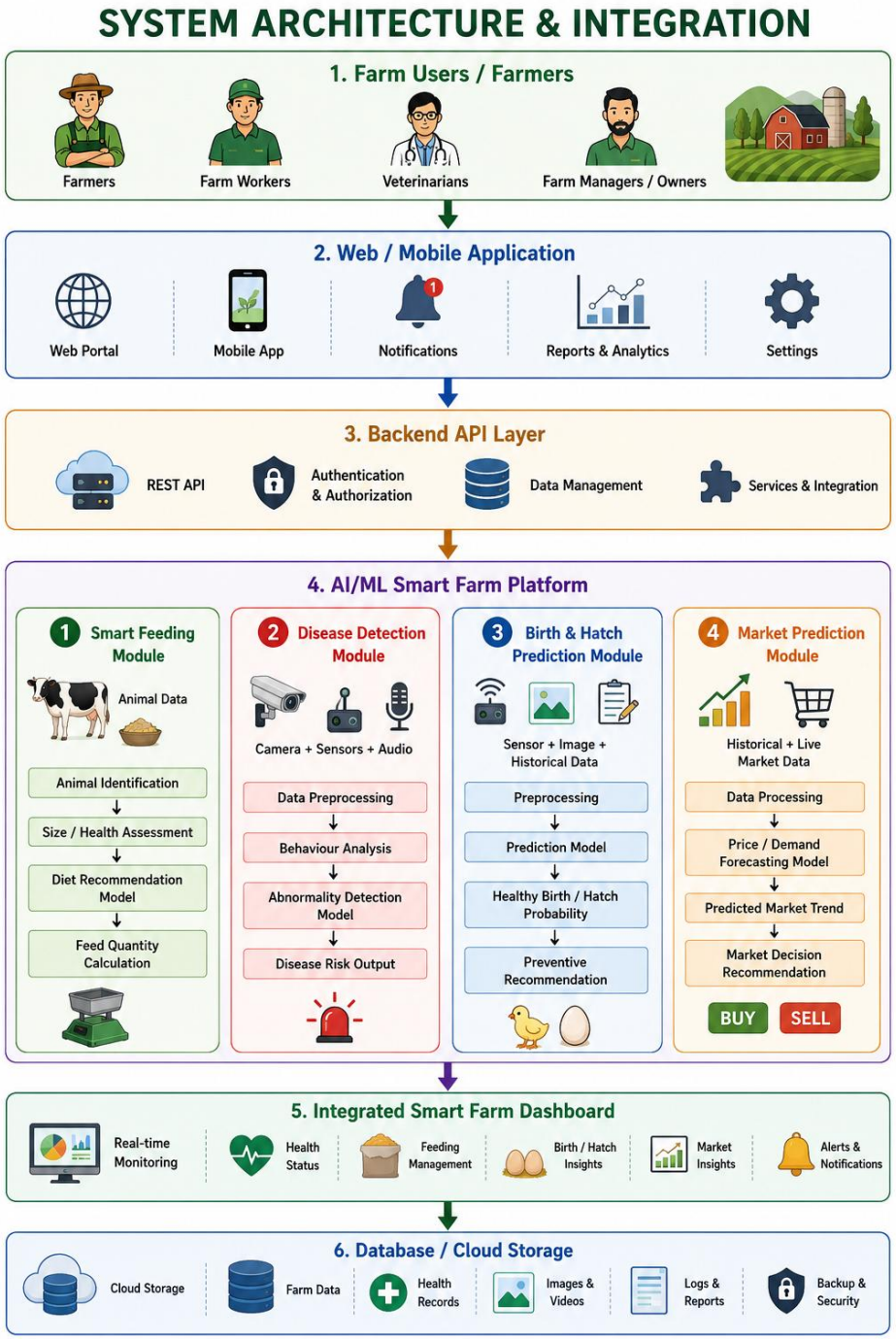
- ❑ Easy-to-use farm management interface
- ❑ Real-time monitoring
- ❑ Fast AI/ML predictions
- ❑ Reliable recommendations
- ❑ Scalable architecture
- ❑ Secure farm data management

Overall Research Gap

Feature	Traditional Farm Management	Existing Single-Purpose Smart Systems	Proposed Smart Farm
Integrated livestock decision support platform	×	×	✓
Real-time animal monitoring	×	✓	✓
Personalized feeding recommendation	×	Partial	✓
Personalized nutrition recommendation	×	Partial	✓
AI-based abnormal behavior detection	×	✓	✓
Disease alerts & preliminary recommendations	×	Partial	✓
Calf birth / egg hatch prediction	×	Partial	✓
Real-time livestock market forecasting	×	Partial	✓
Buy / Sell recommendations	×	Partial	✓
Integration of all four modules	×	×	✓

Research Gap: Existing approaches mainly address individual livestock management problems separately. The proposed solution integrates feeding automation, health monitoring, reproduction prediction, and market forecasting into a single AI/ML-powered smart farm platform.

System Architecture & Integration



Commercialization

Target Users / Market

- Livestock farmers
- Dairy farms
- Poultry farms
- Farm managers and workers
- Veterinarians
- NLDB and other livestock organizations
- Livestock buyers and sellers

Value Proposition

An integrated AI-driven livestock management platform that improves animal nutrition, health monitoring, reproduction management, farm productivity, and market decision-making.

Cost & Feasibility

Can be developed using open-source AI/ML technologies.
Existing farm cameras and sensors can be integrated.
Cloud-based backend can support centralized farm data.
Modular architecture allows gradual implementation.
NLDB farm data can support model development and validation.

Adoption Plan

Phase 1: Model development and testing
Phase 2: Individual module integration
Phase 3: Pilot testing using farm data
Phase 4: Real-world testing in livestock farms
Phase 5: Full smart farm deployment

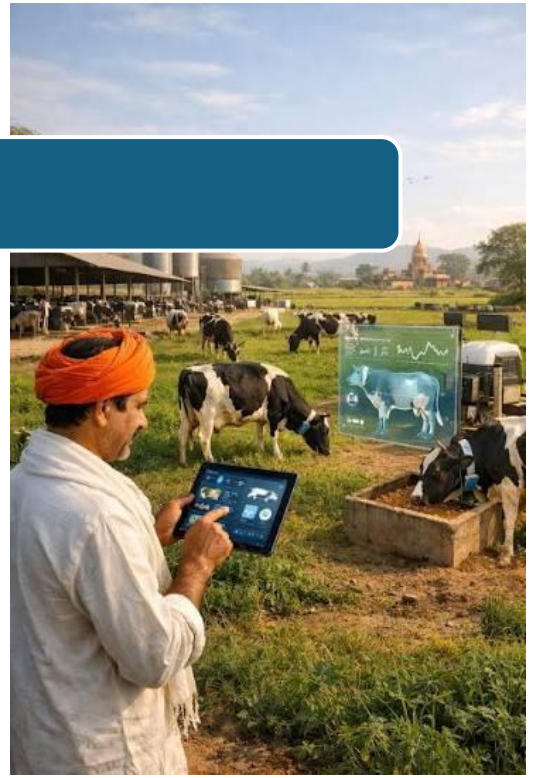


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System Architecture & Integration

Components

- Animal Birth & Egg Hatch Prediction System – **IT22139894**
- Personalized Animal Diet Recommender & Automated Feeding System – **IT22136138**
- Animal Disease Detection and Emergency Alert System – **IT22282118**
- Customized Live Market Prediction & AI Recommendation System – **IT22134608**



Integration Points

- REST API communication
- JSON data exchange
- Image uploads from farm cameras
- Historical animal records
- Feeding and health information
- Market price and demand data
- Shared database
- Centralized dashboard

Expected Impact



- Personalized and balanced animal nutrition
- Reduced feed wastage
- Early detection of health problems
- Faster veterinary intervention
- Reduced disease spread
- Improved healthy birth and egg hatch rates
- Reduced livestock losses
- Better market decisions
- Increased farmer profitability
- Reduced manual workload
- Improved animal welfare
- Data-driven farm management

THANK YOU

